Document Number : 15000
Size: 170mm x190mm

BETAMOX

250mg Tablet & 500mg Tablet

DESCRIPTION

BETAMOX 250MG TABLET: A capsule shape, scored, marked 'USP250' yellow tablet. Each tablet contains Amoxycillin Trihydrate equivallent to Amoxycillin 250mg.
BETAMOX 500MG TABLET: A capsule shape, scored , marked 'UP500' yellow tablet. Each tablet contains Amoxycillin Trihydrate equivalent to Amoxycillin 500mg.

INDICATIONS

Amoxycillin is an aminopenicillin, which has activity against penicillin-sensitive gram-positive bacteria, as well as *Escherichia coli, Proteus mirabilis, Salmonella sp. Shigella sp. and Haemophilus influenza.* However, many *Enterobacteriacaea, H. Influenzae, Salmonella and Shigella* species are resistant to this penicillin because of beta- lactamase production by these organisms Amoxycillin can be administered orally for the following infections caused by susceptible bacterial organisms:

- Bronchitis
- Biliary Tract Infections
- Uncomplicated Endocervical and Urethral Gonorrhoea (caused by susceptible strains of *Neisseria gonorrhoea*)
- Acute Otitis Media
- Mouth Infections
- Pharyngitis
- Pneumonia
- Sinusitis
- Urinary Tract Infections
- Gastro-enteritis (including *E. Coli* enteritis and *Salmonella* enteritis)
- Lyme Disease(early stages monotherapy and later stages in combination with probenecid)
- Typhoid Fever

Amoxycillin is a possible alternative to erythromycin for chlamydial infections in pregnant women. Amoxycillin is also indicated as prophylaxis for spleen disorders (pneumococcal infection) and bacterial endocarditis and adjunct for *Helicobacter pylori* -associated peptic ulcer.

PHARMACOLOGY

Amoxycillin is bactericidal and acts by inhibiting bacterial cell wall synthesis. Its action is dependent on its ability to reach and bind penicillin-binding proteins (PBPs) located on the inner membrane of the bacterial cell wall. PBPs are enzymes responsible for the assembling and reshaping of the bacterial cell wall during growth and division. PBPs inactivation by amoxycillin results in the weakening and lysis of the bacterial cell wall.

PHARMACOKINETICS

Amoxycillin is fairly well absorbed from the GIT. It is more rapidly and more completely absorbed than ampicillin orally. Amoxycillin is acid-stable. Peak plasma concentrations of about 5 microgram per ml have been observed 1-2 hours after a dose of 250mg, with detectable amounts present for up to 8 hours. Doubling the dose can double the concentration. Total amount absorbed is not significantly affected by food. About 20% is plasma protein bound. The plasma half-life is about 1-1.5 hours and this increases (7-20 hours) in renal failure, neonates and the elderly. Amoxycillin is widely distributed at varying concentrations in body tissues and fluids, including peritoneal fluid, blister fluid, pleural fluid, middle ear fluid, intestinal mucosa, bone, gallbladder, urine(high concentrations), lung, female reproductive tissue and bile. It crosses the placenta and small amounts are excreted in breast milk. Little amoxycillin passes into the CSF unless the meninges are inflamed. Amoxycillin is metabolised to a limited extent to penicilloic acid which is excreted in the urine. About 60% of an oral dose is excreted unchanged in the urine in 6 hours by glomerular filtration and tubular secretion; urinary concentrations above 300 microgram per ml have been reported after a dose of 250mg. High concentrations have been reported in bile; some may be excreted in the faeces.

DOSAGE

The usual adult dose is 250-500mg every 8 hours.
Adults:

Severe or recurrent respiratory tract infections	3g twice daily
Uncomplicated endocervical and urethral gonorrhoea (gonococci-sensitive)	Single dose of 3g with probenecid 1g
Dental abscesses/Uncomplicated urinary tract infections	A dose of 3g repeated once after 8 or 10 to 12 hours
Chlamydia(in pregnant women)	500mg every 8 hours for 7 to 10 days
Prophylaxis of endocarditis	3g one hour before dental procedures under local or no anaesthesia then 1.5g 6 hours after the initial dose
Helicobacter pylori associated gastritis or peptic ulcer	500mg 4 times a day or 750mg 3 times a day
Lyme disease	250-500mg 3 or 4 times a day for 3-4 weeks

Children:

Children up to 10 years	125-250mg every 8 hours
Children under 20kg body-weight	20-40mg per kg daily
Otitis media(children 3-10 years)	750mg twice daily for 2 days
Uncomplicated endocervical and urethral gonorrhoea	50mg per kg of body weight and 25mg of probenecid per kg of body weight simultaneously as a single dose in prepubertal children
Lyme disease	6.7-13.3mg per kg of body weight every 8 hours for 10-30 days

Route of Administration: Oral

OVERDOSAGE

There is no specific antidote, treatment of amoxycillin overdose should be symptomatic and supportive. Amoxycillin may be removed from circulation by haemodialysis. Serious hypersensitivity reactions have been reported in patients on penicillin therapy. Serious anaphylactic reactions require immediate emergency treatment with parenteral epinephrine, oxygen, intravenous corticosteroids or air management(including intubation).

For Clostridium diflicile colitis, mild cases may respond to discontinuation of the medication alone. Moderate to severe cases may require fluid, electrolyte, and protein replacement. In cases not responding to the above measures, oral doses of the following may be used:

Vancomycin	125mg every 6 hours for 5-10 days
Metronidazole	250-500mg every 8 hours
Cholestyramine	4g four times a day

If diarrhoea occurs, administration of antiperistaltic antidiarrhoeals (e.g. opioids, diphenoxylate and atropine combination, loperamide, paregoric) is not recommended since they may delay the removal of toxins from the colon, thereby prolonging and/ or worsening the condition.

Up-to-date information on the treatment of overdose can be obtained from The National Poison Centre, University Sains Malaysia.

TOXICOLOGY

Carcinogenicity and mutagenicity:
No documented long-term studies in animals that evaluate the carcinogenic and mutagenic potential of amoxycillin were available.
Pregnanty/Reproduction:
(FDA pregnancy category: B)
Studies in mice and rats at doses up to 10 times the human dose of amoxycillin revealed no evidence of impaired fertility. Amoxycillin crosses the placenta. Adequate and well-controlled studies in humans have not been done to determine whether penicillins are teratogenic; however, penicillins are widely used in pregnant women and problems have not been documented to date. Studies in mice and rats at doses up to 10 times the human dose revealed no evidence of harm to the foetus.
Breast feeding:
Amoxycillin is distributed into breast milk, usually in low concentration. Although significant problems in humans have not been documented, the use of amoxycillin by nursing mothers may lead to sensitization, diarrhoea, candidiasis, and skin rash in the infant.
Pediatrics:
Penicillins, including amoxycillin have been used in pediatric patients and no pediatrics-specific problems have been documented to date. However, incompletely developed renal function of neonates and young infants may delay the excretion of renally eliminated penicillins.
Geriatrics:
Penicillins, including amoxycillin have been used in geriatric patients and no geriatrics-specific problems have been documented to date. However, elderly patients are more likely to have age-related renal function impairment, which may require an adjustment in dosage in patients receiving penicillins.
Dental:
Prolonged use of amoxycillin may lead to the development of oral candidiasis.

SIDE EFFECTS

Minor side effects like GIT reactions (mild diarrhoea, nausea or vomiting) may occasionally occur. There have also been reports of headache, oral candidiasis(sore mouth or tongue, white patches in mouth and/ or tongue), vaginal candidiasis(vaginal itching and discharge) and pseudomembranous colitis.
Allergic reactions, specifically anaphylaxis(fast or irregular breathing; puffiness around the face; shortness of breath; sudden, severe decrease in blood pressure), exfoliative dermatitis(red, scaly skin), skin rash, hives, itching and serum sickness-like reactions(skin rash, joint pain, fever) are uncommon adverse reactions.
There have been rare reports of hepatotoxicity(fever, nausea and vomiting, yellow eyes or skin); interstitial nephritis(fever, possibly decreased urine output, skin rash); leucopenia or neutropenia(sore throat and fever); mental disturbances(anxiety, confusion, agitation or combativeness, depression, seizures, hallucinations, or expressed fear of impending death); platelet dysfunction or thrombocytopenia(unusual bleeding or bruising); *Clostridium difficile* colitis or seizure.
Skin and subcutaneous tissue disorders: Frequency 'very rare': Drug Reaction with Eosinophillia and Systemic Symptoms (DRESS).
Effects on Ability to Drive and Use Machines:
No studies on the effects on the ability to drive and use machines have been performed.
However, undesirable effects may occur (e.g. allergic reactions, dizziness, convulsions), which may influence the ability to drive and use machines.

CONTRAINDICATIONS

Betamox is contraindicated in patients with hypersensitivity to penicillins. Risk-benefit should be considered when amoxycillin is administered in the presence of bleeding disorders, GI disease, infectious mononucleosis and renal function impairment.

DRUG INTERACTIONS

There have been case reports of reduced oral contraceptive effectiveness, resulting in unplanned pregnancy. An alternative or additional method of contraceptive should be advised.
An increased frequency of skin rashes has been reported in patients receiving amoxycillin together with allopurinol, but this is yet to be confirmed.
Probenecid decrease renal tubular secretion of amoxycillin when used concurrently and may increase the risk of toxicity.
Amoxycillin may decrease the clearance of methotrexate and may cause methotrexate toxicity.
Other drugs which interaction may occur:
Angiotensin-converting enzyme(ACE) inhibitors, potassium-sparing diuretics, potassium supplements.

WARNING & PRECAUTIONS

Before initiating therapy with Betamox, careful inquiry should be made concerning previous hypersensitivity reaction to penicillins, cephalosporins or other allergens. If an allergic reaction occurs, Betamox should be discontinued and appropriate therapy instituted. Caution should be practised for use in children, or if given concurrently with other medication or the existence of other medical problems.
High urinary concentrations of a penicillin may produce false-positive or falsely elevated glucose in the urine with copper sulfate test; glucose enzymatic tests are not affected. False-positive result may occur during therapy with any penicillin in the Direct antiglobulin (Coombs') test.
Physiology/laboratory test values of Alanine aminotransferase (ALT[SGPT]), Alkaline phosphatase, Aspartate aminotransferase (AST[SGOT]) and serum Lactate dehydrogenase (LDH) may be increased.
Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients on penicillin therapy.
Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Betamox, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, Betamox must be discontinued immediately and appropriate alternative therapy instituted.

PRESENTATION

Tablet 250mg 10s, 10 x 10's, 50 x 10's and 100 x 10s. (Blister pack) & 100's. (For Export)
Tablet 500mg 10s, 2 x 14's, 4 x 14's, 10 x 10's, 50 x 10's and 100 x 10s. (Blister pack)

STORAGE CONDITIONS

Store in a dry place below 30°C
Protect from light
Keep out of reach of children
Jauhi daripada kanak-kanak
Shelf life: Please refer to outer package.
Take at regular intervals. Complete the prescribed course unless otherwise directed.
May be taken on a full or empty stomach.
May be taken with baby formulas, milk, fruit juice, water, or other cold drinks.

Product Registration Holder & Manufacturer:
Duopharma (M) Sdn. Bhd.
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DUOPHARMA

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Revision Date: 14 January 2022